Cofinal anchors and height one equations for ZFC consistency

Round 17 research report and reproducible artifacts

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Abstract

This report gives explicit polynomial equations whose unsolvability is equivalent to syntactic ZFC consistency, relative to the identified inherited ordinary calibrations. A cofinal semantic Pell anchor and strict quotient bounds yield natural-domain quartics with 229 or 230 variables; a 228-variable companion trades a much larger coefficient height for one fewer coordinate. A separate signed construction removes front sign guards only after an orientation-first recovery argument, yielding quartics in 282, 287 or 309 unrestricted integers with coefficient height 128446. Incidence-local alias contraction, integral affine refactorization and a quartic two-alias equality polynomial give height-one quartics in 793, 806 or 994 unrestricted integers, with explicit bijective witness maps. Finally, a linear cofactor saturation gives eleven-integer equations at exact degree and operation pairs 328/130, 340/129, 352/125 and 398/121. These are different domain-specific Pareto objectives, not a single simplicity optimum. Complete new unbounded proofs, literal row and coefficient arrays, arithmetic graphs, authentic numerical ports, deterministic builders and independent finite checkers accompany the report. Neither consistency nor a new end-to-end proof-assistant endpoint is asserted.

Contents

1 Results and their precise interpretation **4**
1.1 The exact logical statement 5

2 Literal polynomial conventions **5**
2.1 The complete coherent coefficient tuple 6
2.2 Reading the proof text 6

3 One semantic Pell anchor and an aligned amplification index **6**
3.1 Result and meaning of the equation 6
3.2 A general one-row semantic-anchor lemma 7
3.3 The literal semantic frame and anchor certificate 8
3.4 Power recovery and modular numerators 8
3.5 Bootstrap and factorial/product fusion 9
3.6 Why x is enough amplification 10
3.7 The self-sized radix and CRT soundness 10
3.8 Constructive completeness and anchor dominance 11
3.9 Consecutive coordinates and exact contraction 11
3.10 Fixed-priority 228-variable companion 11

4 Reusing a semantic Pell anchor over unrestricted integers **12**
4.1 Result and scope 12
4.2 Shared notation and unchanged front 12
4.3 The reused signed anchor 13
4.4 Pooled signed-index recovery including the anchor 13
4.5 Powers and the acyclic front-sign schedule 14
4.6 Cofinal semantic-anchor completeness 14
4.7 Literal compilation and audit boundary 15

5 Recovering nonnegative arithmetic from signed Pell orientation **15**
5.1 The hypotheses available before any semantic arithmetic 15
5.2 A signed nominal index in the squared anchor 16
5.3 Both coordinate signs from bounded errors 16
5.4 Negative requested exponents are impossible at a positive base 16
5.5 The complete acyclic recovery schedule 17
5.6 Zero representatives and the precise semantic boundary 18
5.7 Completeness and exact audit 18

6 Aligning the amplification exponent with a strict beta quotient range **19**
6.1 Exact changes to the arithmetic interface 19
6.2 Noncircular power and endpoint recovery 19
6.3 Amplification, precision and radix bounds 19
6.4 Constructive completeness for the strict ranges 20
6.5 Sparse coordinate residues 20
6.6 Conclusion and verification boundary 21

7 Linear cofactor saturation in eleven unrestricted integers **21**
7.1 Result and mathematical boundary 21
7.2 1. Complete ordinary polynomials 21
7.3 2. Unconditional arithmetic and the guard 22
7.4 3. Two squareclasses and the linear cofactor 23
7.5 4. Integral auxiliary root and carrier divisibility 24

7.6 5. Scale restoration and exact native source recovery 24
7.7 6. Converse with one fixed strong pair 26
7.8 7. Exact degrees and actual leading coefficients 26
7.9 8. Portable checks and inherited dependencies 27

8 Height one compilation and its exact witness maps 28
8.1 The three fixed height one equations 28
8.2 The result and its scope 28
8.3 Exact affine refactorization 29
8.4 Unit coefficient normalization 29
8.5 A shared incidence graph 29
8.6 Incidence local contraction 30
8.7 Equality and unit penalties 30
8.8 Explicit forward witness map 31
8.9 Explicit inverse and uniqueness 31
8.10 Reproducibility and evidence boundaries 32

9 Comparison with the pinned public collection 32
9.1 Natural quartic comparison 32
9.2 Signed direct quartic comparison 33
9.3 Height one and small-variable objectives 33
9.4 Attribution and limits 34

10 Reproduction and evidence boundaries 34
10.1 Portable proof closure 35
10.2 Report fidelity 35

11 Unfinished directions 35

1 Results and their precise interpretation

Write $\mathbb{N} = \{0, 1, 2, \ldots\}$ and let $\mathbb{Z}$ denote unrestricted integers. Domains are part of each theorem and are never silently interchanged. Every independent coordinate is counted; an arithmetic abbreviation does not introduce another quantified variable. Coefficient height is measured after full collection, whereas residual rows and live binary operations are different quantities.

Natural quartics over $\mathbb{N}$

Endpoint	Variables	Degree	Rows	Terms	Height
Strict cofinal anchor	229	4	179	9184	83954634
Sparse cofinal anchor	230	4	180	6858	83954472
High-height projection	228	4	178	9866	2445412949744

Direct quartics over $\mathbb{Z}$

Endpoint	Variables	Degree	Rows	Terms	Height
Signed dimension	282	4	192	5491	128446
Signed grouped	287	4	197	4313	128446
Signed sparse	309	4	219	3922	128446

Height one quartics over $\mathbb{Z}$

Endpoint	Variables	Degree	Terms	Height	Constant
Height one compact	793	4	51982	1	1
Height one balanced	806	4	23719	1	1
Height one sparse	994	4	11744	1	1

Eleven integer arithmetic graphs

Endpoint	Variables	Degree	Operations
Sharp free cofactor	11	328	130
Simple free cofactor	11	340	129
Uncentered free cofactor	11	352	125
Uncentered weak cofactor	11	398	121

The natural cofinal-anchor family uses a single bounded-error row both to bound actual Pell indices and to match them to the requested indices. Aligning the amplification exponent with x is justified by strict beta-quotient bounds, not by substituting x into a theorem proved only for a larger exponent. Its smaller 228-variable companion has coefficient height 2445412949744, so it is not the preferred balanced equation.

The signed family has its own complete sign proof. Orientation of the Pell pairs is established before deriving the signs of the exposed encoded representatives and quotient endpoint. This is not an unrestricted-domain reinterpretation of the natural certificate. Its witnesses are reconstructed with the cofinal representative and anchor choice in an explicit noncircular order.

The height-one family optimizes a different axis. It has more variables than the direct signed equations; the gain is that every collected nonzero coefficient is 1 or -1. Actual incidence support, including harmless signed cancellations, replaces a stronger global collision prohibition. The independent check expands the whole final polynomial, so this improvement does not rely on an unverified graph heuristic.

For the eleven-variable family, linear cofactor saturation and the sharp polynomial center lower degree while preserving a complete authentic fixed coefficient tuple. The four recommended

graphs are mutually nondominated in the reported degree/cost pair. Their actual highest-degree coefficients are nonzero at that tuple, not merely for generic coefficient symbols. Lower-cost higher-degree predecessors are retained as comparison points, not misclassified as dominated.

1.1 The exact logical statement

For each fixed equation P on its stated domain D, the new ordinary arguments and the identified inherited calibration give

$$(\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0) \quad \Longleftrightarrow \quad \text{there exists a finite syntactic ZFC refutation.}$$

Consequently the claimed object-theoretic statement is

$$\mathrm{ZFC} \vdash \left[\forall \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) \neq 0\right] \Longleftrightarrow \mathrm{Con(ZFC)}.$$

Neither side is asserted to be true by this report. The finite checkers prove no consistency statement and do not decide the zero sets. No fresh transitive Lean or Rocq kernel build, encoded object-ZFC derivation, global minimum, or literature-wide novelty claim is made.

The direct logical interfaces are the retained named ordinary statements

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The scalar route retains its source34/native/compiler/refutation theorem and the specifically identified signed-rank and interpolation arguments. The supplied numerical and executable closure is local, but this is not a new self-contained reconstruction of every historical native/compiler theorem. Inherited Lean files document the boundary; their inclusion alone is not a formal certification of the new equations.

2 Literal polynomial conventions

The JSON files are mathematical definitions, not merely summaries of an implementation. For each natural or signed direct endpoint, its `equation.json` ordered list `variables` fixes the independent coordinates $x_0, \ldots, x_{n-1}$. A term `[c,[i,j]]` means cx_ix_j, a repeated index gives a power, and an empty index list means the constant c. Every integer coefficient is stored in full. A residual row is the sum of its listed terms. The equation is exactly

$$P(x_0, \ldots, x_{n-1}) = \sum_{r=0}^{m-1} R_r(x)^2 = 0.$$

The separately supplied `expanded_polynomial.json` is the collected sum, including its constant coefficient. In that expanded SOS file, `variables` stores the coordinate count; the ordered names remain in the adjacent equation file. Thus the displayed definition has no unbounded product, variable exponent, factorial, binomial, unspecified numeral, or quantifier as an algebraic primitive. Those operations occur only in the mathematical interpretation of zeros. The final rows have degree at most two, and the supplied nonzero quartic coefficients prove the exact final degree, rather than only an upper bound.

The height-one endpoints instead store the final collected polynomial directly. Their certificates list every vertex, edge, factor occurrence, equality class and unit path. The report proves the exact half-sum-of-squares identity of this polynomial, and the independent checker recollects that identity. It does not replace the published coefficient file by an informal graph specification. All nonzero coefficients are exactly 1 or -1, the constant coefficient is exactly 1, and every counted variable occurs.

For the eleven-integer equations, the complete acyclic arithmetic graph is the ordinary polynomial definition. A leaf is an integer constant, one of eleven independent integer variables, or one

of six *fixed* numerical ports. Each internal gate is binary addition, subtraction or multiplication. The output recursively denotes a unique element of $\mathbb{Z}[j, C, z, h, r, v, w, T, c, n, s]$. Named registers are abbreviations, not new unknowns. There is no division or radical gate. Cost counts live binary operations, including squares and multiplication by numerical constants; copying a register and reading a fixed leaf are free.

2.1 The complete coherent coefficient tuple

The scalar family uses only the six-port source34 tuple named `mixed-parity-unrounded-1064921`. Its source commit is

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

Its constants are $\beta = 2^{1064921} - 1$, cone_A $= 2^{1113}$, cone_B $= 2^{666104}$ and the literal K_0, M_C, M_F payloads. The M_F payload is already the native value plus β; it must not be shifted again. The binding records the uncompressed byte length, bit length, endianness and SHA-256 for every port. The compressed files contain the complete minimal unsigned little-endian integers, not shortened decimal displays, hash-only stand-ins or symbolic placeholders. The full finite compiler recipe reconstructs the same payloads from the included machine and tile data.

No coefficient from the compact44 positive-domain tuple or another historical source is substituted into this tuple. Completeness and soundness are claims for the fixed authentic numerals. The large numerical values affect coefficient height but do not add existential variables. For the scalar graphs the exact degree and operation count are certified, with an explicitly attained highest-degree coefficient; their full collected support and overall coefficient height are *not* claimed to have been materialized.

2.2 Reading the proof text

The new natural, signed and scalar proofs below are included unabridged. Literal ASCII formulas preserve exponent carets, inequalities and subscripts exactly, with presentation-only line wrapping. The independent character checker removes Markdown and TeX presentation separately and compares every remaining non-whitespace character. The height-one proof is native mathematical TeX; its exact source is separately retained. Complete inherited proof sources are included in the archive, with an active dependency map rather than an assertion that every historical file is part of the new result.

3 One semantic Pell anchor and an aligned amplification index

3.1 Result and meaning of the equation

The main new natural-domain construction is the sum of squares of the 179 quadratic integer residuals in `strict_natural229/natural/equation.json`. It has 229 nonnegative integer variables. Its independently collected quartic has 9,184 nonzero monomials and coefficient height 83,954,634. The sparse companion `strict_natural230` has 230 variables, 180 rows, 6,858 monomials and height 83,954,472. An additional exact-projection companion has 228 variables, 178 rows, 9,866 monomials and height 2,445,412,949,744. It trades much greater coefficient height for one fewer variable and is not the recommended balanced form. Each coefficient is supplied literally in the respective `expanded_polynomial.json`; no exponentiation, factorial, binomial, bounded quantifier, or implicit sequence is a primitive of that polynomial.

The existence claim is unchanged: with the inherited ordinary numerical C222 calculus calibration, the polynomial has a natural zero exactly when ZFC has a finite syntactic refutation. Consequently ZFC proves that its no-zero sentence is equivalent to Con(ZFC). This is neither a proof of consistency, a global minimality assertion, nor a newly checked end-to-end proof-assistant or object-language ZFC derivation.

There are two new compatible reductions. First, a single bounded-error row reuses the already required derived-label Pell index as the full anchor. Its index need not be even and the size frame need not be doubled. Second, the amplification exponent is the largest required quotient-range index x, rather than a separate large affine expression. The two beta quotient ranges are sharpened from u,v<=x to u,v<x, as their defining equalities already permit. The incoming factorial/ product fusion, modular binomial graphs, dominant denominator, consecutive Pell coordinates, and self-sized radix are explicitly retained and credited.

3.2 A general one-row semantic-anchor lemma

Write X_j(A),Y_j(A) for the ordinary Pell sequences defined by

```
(A+sqrt(A^2-1))^j = X_j(A)+Y_j(A)sqrt(A^2-1).
```

Assume integers W>=3, 2<=H<W, A-1>W^3, r<=W-2 and requested indices 1<=k_i<W. Retain a complete natural indexed-Pell certificate at index H, with principal pair (U,V), omitting only its separate inequality H<=V. For each i retain a natural principal Pell norm. Impose the single row

```
V = H + sum_i (Y_i-k_i)^2 + (A-1)Q,        Q>=0.              (1)
```

Then the principal pair is (X_(k_i)(A),Y_(k_i)(A)) for every i. Conversely, canonical pairs extend to (1) whenever H>=2 max_i k_i. All private witnesses of the complete indexed-Pell certificate retain their inherited natural domains. The assertion does not license making any private witness signed.

For soundness, (1) supplies V>=H, so the complete anchor theorem first gives V=Y_H(A). Since H>=2, V>=Y_2(A)=2A>2W. Classification of each principal norm gives Y_i=Y_(j_ i)(A), j_i>=0. If j_i>=H, then Y_i>=V and

```
(Y_i-k_i)^2 >= (V-W+1)^2 > V.
```

For the last strict inequality, V>=2W gives V-W+1>=V/2+1, whose square is larger than V. This contradicts (1), which bounds the sum of error squares by V-H<V. Hence j_i<H<W. The Pell recurrence modulo A-1 gives Y_j(A)=j modulo A-1. Reducing (1) therefore gives

```
A-1 divides E = sum_i (j_i-k_i)^2,
0<=E<=r(W-1)^2<W^3<A-1.
```

Thus E=0 and j_i=k_i. In particular, no separate hypothesis k_i<H was used in soundness. If ordinary coordinates are used, natural X_i supplies the positive abscissa. Consecutive coordinates are handled below.

For completeness, put s=max_i k_i>=1 and z=Y_s(A)>=1. If H>=2s, then

```
V=Y_H(A)>=Y_(2s)(A)=2X_s(A)Y_s(A)>2(A-1)z^2.
```

As Y_(k_i)(A)>=k_i, each actual square error is at most z^2. Therefore

```
V-sum_i(Y_(k_i)(A)-k_i)^2 > [2(A-1)-r]z^2 > H,
```

where A-1>W^3, r<=W-2 and H<W imply 2(A-1)-r>H. The numerator V-H-sum_i(Y_(k_ i)(A)-k_i)^2 is positive and divisible by A-1. Its quotient is the required natural Q. The inherited indexed-Pell completeness theorem supplies fresh private anchor witnesses at this A and H. This proof uses neither parity of H nor a doubled frame.

3.3 The literal semantic frame and anchor certificate

Let t=t0+12 and take the unchanged positive encoded representatives C,A_a,A_b,A_c,A_d,A_S,A_Sp,A_u,A_v. The derived nonnegative label polynomial J is the sum of the nine axiom conclusion polynomials and a,b,I(a,b),A(a,b),1, with the same Cantor constructors and arithmetic source as the predecessor. In particular J>=I(A_a,A_b)>=2A_a^2>=2. Use H=J in (1).

The frame is

```
W = 15 + sum(distinct exact-power bases, exponents and outputs
             together with all modular numerator indices).
```

There are thirteen independent seeds; J itself uses the full anchor pair and is not also a seed. Thus H<W, every intended index is <W and r=13<=W-2. The current code computes the offset as the actual seed count plus two, rather than trusting an unattested hard-coded census.

The scale rows are

```
L^2-((W+1)^2-1)(WZ)^2=1,    Z=z+1,    A=L+sigma,
```

with all coordinates natural. Pell classification gives L=X_e(W+1), WZ=Y_e(W+1) with e>0. Reduction modulo W gives W|e, so e>=W and A>=L>=X_W(W+1)>=(W+1)^W. In particular A-1>W^3 and A>W+1, before interpreting any exact power. The free sigma permits choosing A in the later common-root congruence class.

The full anchor is exactly the inherited three-norm indexed-Pell graph. Using private names U1,V1,B1,S1,R1,d,e,f,g,h, it is

```
U^2-(A^2-1)V^2=1,
U1^2-(A^2-1)V1^2=1,
S1^2-(B1^2-1)R1^2=1,
B1=1+4V(d+1),                  V1=V^2(e+1),
B1=A+U1*f,                    S1=U+U1*g,
R1=H+4V*h.
```

Its omitted H<=V condition follows from (1). The existing constructive criterion also supplies the nonnegative oriented congruence quotients: H>=2 gives V>=2A; B1>1 and B1=1 modulo 4V give B1>A; R1>=H>=2 and Pell classification at B1 give R1>=2B1, hence S1>U. No unsupported new indexed-Pell criterion is being substituted.

3.4 Power recovery and modular numerators

Each exact request m=n^k retains

```
T_n=2An-n^2-1,
X_k(A)+nY_k(A)=A Y_k(A)+m+T_n*q_n,
```

with T_n,q_n natural. The first row excludes n=0. Every requested exponent is positive from the raw expressions, endpoints and positive representative offsets. After (1), the ordinary Pell congruence gives m=n^k modulo T_n. The frame bounds n,k,m by W, while A>=(W+1)^W gives T_n>W^W+W. Thus the congruence is an equality. Canonical positive powers provide nonnegative quotients by the inherited positive-power argument.

The six exact requests are

```
V0=(F^2+1)^t,       N=B^K,           q=(N*S)^K,
D_t=q^t,           D_y=q^(y+1),     D_x=q^x,
K=x,             S=1+C+sum(eight encoded columns).
```

The extra denominator D_(t+1)=q∗D_t is multiplication, not a seventh exact power. Aligning K with x makes two formerly distinct index requests literally the same atom. All eleven modular numerator indices remain in W.

The common-root equation is

```
2A(q+1)=(q+1)^2+1+M*h0,        M=N*D_x.                    (2)
```

Once the six powers and endpoints are decoded, x>t+1 and x>y+1; hence M is divisible by qD_t for the one exact binomial, and by N times every range denominator. Substituting s=q+1-A into the Pell identity modulo M gives

```
X_n(A)+(q+1-A)Y_n(A) = (q+1)^n modulo M.                (3)
```

The left side is positive: X_n(A)>(A-1)Y_n(A) for n>0, and every n here is positive. The literal modular digit equations are the same as the incoming common-root compiler, with Euclidean remainder r<q^k and, for the exact binomial only, output B<q. Thus they recover exactly the desired low digit or binomial divisibility once the carry bounds below hold.

For completeness, N|q, so all prime divisors of M divide q and gcd(q+1,M)=1. Hence gcd(2(q+1),M) is one or two. In the latter case q+1 is odd and (q+1)^2+1 is even, so the linear congruence (2) is soluble. Choose A in its solution class arbitrarily large after all semantic data are fixed; both sigma and h0 can be natural. Canonical Pell residues are positive and congruent to their least nonnegative required residues, so all modular digit quotients are natural.

3.5 Bootstrap and factorial/product fusion

The factorial/product output B is shared by the two ceiling rows

```
V0+B=B*F+u+1,        B=u+v+1.
```

They imply B>=1 and V0+v=BF. Since V0=(F^2+1)^t>=1, F>=1. Also V0>F^2, so B>F and V0<=BF<B^2. The raw endpoint x>=F+1 gives K=x>=2. Consequently

```
V0<B^2<=N<=q.
```

For r>=1 and n>=1, every coefficient binom(r+n,j), 0<=j<=n, is at most (r+1)^n. This follows by Pascal induction for j<n and the product formula for j=n. Apply this with r=F^2,n=t. All coefficients through t are <=V0<q, so the exact modular digit really gives B=binom(F^2+t,t), without first assuming the factorial identity.

The ceiling rows say F=ceil(V0/B). Write T=t! and

```
V0/B=T*product_(i=1)^t (F^2+1)/(F^2+i).
```

Every factor is <=1, so F<=T. Monotonicity in F gives V0/B>=2^t/(t+1)>t^2 for t>=12, hence F>t^2. The product bound product(1-a_i)>=1-sum a_i with a_i=(i-1)/(F^2+i) gives

```
T*(1-1/(2F))<V0/B<=F.
```

Thus T<F+F/(2F-1)<=F+1, so T<=F by integrality. Therefore F=t!. Conversely at F=T the same estimate gives T-1/2<V0/B<=T, furnishing natural ceiling witnesses. This is the inherited square-radius identity.

For 1<=i<=t, d_i=1+F^2/i is integral and congruent to one modulo F. Any prime divisor of d_i exceeds t, and the d_i are pairwise coprime, because a common prime for distinct indices would divide their difference. Their product is precisely B. This recovers the reciprocal CRT carrier.

3.6 Why x is enough amplification

Retain the literal source majorant with E=2^23 and base R=2+t+x+y+F, so each nonnegative local source value and local label sum is strictly below R^E at the prescribed local bounds. The inherited J majorant has ell=2^22.

After F=t!, the endpoint equations give

```
y>=F+1,
x=1+(1+(t+1)y)*u_t,      x>=F+1>=2.
```

Thus u_t>=1 and x>=2+(t+1)y. For t>=12, t<y and F<y, so

```
R<2+x+3y<2x,
x>=2+13*(12!+1)>2^32.
```

At r0=2^32, (2r0)^E=2^(33E)<2^r0 because 33E=276,824,064<2^32. For r>=r0, the binomial theorem and binom(E,j)<=E^j give

```
(1+1/r)^E <= sum_(j=0)^E (E/r)^j
             < 1/(1-E/r) < 2,
```

since E/r<=1/512. Induction therefore proves (2r)^E<2^r for all integers r>=r0. In particular every required local value is

```
<R^E<(2x)^E<2^x=2^K.
```

This proof does not assume a bound on x in terms of y; the arbitrarily large beta representative x makes the estimate stronger, not weaker. For the only tightened ranges, the inherited beta equalities in a canonical finite-refutation witness have u=(x-S)/(1+(i+1)y) and v=(x-Sp)/(1+(i+2)y), with S,Sp>=0, y>=1 and x>0. Each denominator is at least two, so u,v<=x/2<x. Thus completeness admits lower index x instead of x+1 for these two ranges. Soundness only strengthens the old bounds; all local values still lie in the old majorant's permitted box.

All lower range indices t,t+1,y+1,x are <=K. Equality for x is permitted: in the completeness valuation estimate division by k! loses at most k-1<=K-1, which is still strictly less than K.

3.7 The self-sized radix and CRT soundness

Every independently chosen positive representative is <S. For every such representative n and every 0<=j<=k<=K,

```
binom(n,j)<=max(1,n^K)<S^K<q.
```

The coefficients with j>n are zero. No aggregate upper-bound equation or slack is required. The literal majorant gives J<=S^ell. At t=12, 3ell(t+1)<2∗12!, and this inequality persists with t by factorial growth. Hence ell(t+1)<F<K, and every required coefficient of J through index t+1 is also <S^K<q. These bounds apply to arbitrary representatives in soundness, including the representative deliberately enlarged below.

The modular graphs therefore give all ten intended binomial divisibilities by N. The inherited prime-power lemma says that p^e|binom(a,k), k,e>=1, forces a modulo p^e to lie in [0,k-1]. It is used only in that direction. For row i-1 choose p|d_i and m=v_p(d_i). The raw B divisibility of C+F^2+1 identifies C modulo p as i-1. Its range then identifies the small residue modulo p^(Km) as exactly that index. The remaining ranges recover every local coordinate in its prescribed interval. The local source divisibility by N, together with its strict bound <2^K<=p^(Km), forces the nonnegative local source value to equal zero. The derived-label range similarly forces J(local)<=t. The unchanged beta consistency and pairwise-coprime weight/Horn-flow argument then produce a finite C222 refutation.

3.8 Constructive completeness and anchor dominance

Start with a finite refutation, expand to a proof tree and pad a horizon t>=12 beyond every required local parameter and label sum. The inherited beta construction supplies x,y and the local source. Set F=t!, B, K=x and N=B^K. Choose each of the nine positive independent representatives by CRT modulo d_i^(2K), initially in (0,N^2]. Replace only A_a by A_a+N^2. This preserves every local polynomial congruence to precision 2K, and leaves every representative <=2N^2.

For each needed range, the falling-factorial numerator contains a term with at least 2Km p-adic valuation, while division by k! loses at most k-1<=K-1. The result therefore has at least Km valuation. Polynomial congruence gives the same conclusion for J. This is the genuine precision-2K completeness argument, not the false converse of the range lemma.

Let R=2+t+x+y+F. The endpoint inequalities above give R<2x. More directly, F,y,t<x, so the remaining nominal indices t,y+1,x,F^2+t are <=x^2+x. For x>=5, x^2+x<2^x, by the base case x=5 and induction. Since N=B^x>=2^x, every such index is <N. All other nonanchor indices are representatives <=2N^2. Thus s=max_i k_i<=2N^2. But

```
H=J>=2(A_a+N^2)^2>2N^4>4N^2>=2s,
```

as N>=2. This supplies the extra completeness hypothesis of (1), at no additional witness or row cost. Compute S and q afterwards; their carry bounds hold regardless of the actual representatives' sizes. Then compute the exact denominators, W, L and a sufficiently large A satisfying (2). Supply canonical Pell pairs, fresh full-anchor witnesses, Q from (1), and Euclidean modular-digit witnesses. There is no cyclic choice involving q, A, the anchor, or the CRT representatives.

3.9 Consecutive coordinates and exact contraction

For every nonanchor seed the chosen endpoint stores natural Y,Z and a product wire P=YZ, and imposes

```
Y^2+Z^2-2A*P=1,        nY=Z+R_n,
```

where n is an exact-power base or q+1 and R_n is the nonnegative sum in its actual output row. Each seed is checked to have such an incidence. The frame gives 1<=n<A. The output row excludes Y=0 because the norm would force Z=1. It gives Z<=nY<AY, so X=AY-Z>0. Expanding the conic gives X^2-(A^2-1)Y^2=1; positive Pell classification follows. Thus (1) applies to its ordinate, and canonical Z=Y_(k-1)(A) supplies completeness. The full anchor retains ordinary positive coordinates and every private witness. No anchor norm is silently replaced by an unindexed seed.

Finally the raw polynomial graph is exported and contracted only through explicit integer polynomial restoration certificates. Their nonnegativity is witnessed by rational sums of affine squares plus nonnegative monomials. The independent checker expands these identities, replays each substitution, checks that rows stay quadratic, and independently collects the complete sum of squares. Thus both directions of natural existence are preserved. Finite arithmetic tests and syntax checks complement the ordinary unbounded proof above; they do not replace it or the inherited logical calibration.

3.10 Fixed-priority 228-variable companion

The 228-variable companion starts from the identical raw semantic equation. Its contractor uses a fixed, explicitly listed order of row priorities, bound to the raw equation's canonical SHA-256, rather than the predecessor's ascending row order. Every elimination is still an integer monic substitution certified nonnegative by exactly expanded rational squares and nonnegative

monomials. The final certificate has 106 eliminations, including two restorations with negative expanded coefficients; both have independently checked positive-square certificates. The target has 228 variables, 178 residuals, 9,866 quartic terms and coefficient height 2,445,412,949,744. No semantic inference or domain relaxation is used for this extra variable reduction. Independent normal and optimized runs replay every restoration and all collected coefficients.

4 Reusing a semantic Pell anchor over unrestricted integers

4.1 Result and scope

The selected exact equation is the sum of the squares of the 204 quadratic residuals in `integer315/equation.json`. Its 315 variables range over all integers. Its separately materialized collected polynomial has degree four, 5709 nonzero monomials and coefficient height 128446. The conservative comparison `integer321` has 321 variables, 206 rows, 5785 monomials and the same height. These are literal finite counts, not global minimality claims.

The input is precisely the round16 direct portable source. The two input files are bound by SHA256 in the independent checker. All eighteen power requests, the square-output factorial fixed point, the cofinal horizon, the self-sized radix, the range tests and the finite Horn-refutation calibration are unchanged. The inherited semantic boundary is the ordinary interpretation associated with `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. We do not claim a new end-to-end proof-assistant certificate or a proof of consistency. The result is an ordinary mathematical proof, formalizable in ZFC, that integer solvability is equivalent to the existence of a finite ZFC refutation.

The round16 `direct/PROOF.md` supplies the complete arithmetic-front argument. This note gives the replacement private Pell certificate, its full sign schedule, and the changed completeness choice. In particular it does not silently treat a signed private coordinate as a natural one.

4.2 Shared notation and unchanged front

Write X_j(a), Y_j(a) for the Pell polynomials defined by X_0=1, Y_0=0 and

```
X_(j+1)=a X_j+(a^2-1)Y_j,
Y_(j+1)=X_j+aY_j.
```

For j>0 extend to negative indices by X_(-j)=X_j and Y_(-j)=-Y_j. For negative centers,

```
X_j(-a)=(-1)^j X_j(a),
Y_j(-a)=(-1)^(j-1)Y_j(a)  (j>0).
```

The unchanged square frame W=2+sum(atom^2) includes every actual affine base, exponent and output atom of all eighteen requests. Thus W>=258, |atom|<W, atom^2<W, and every requested absolute index is less than W. The fifteen principal pairs include the already needed derived-label pair. Its requested index is denoted h; literally h=1+add.380. It is a monotone syntax-constructor expression in the four positive CRT representatives, so h>=2 before interpreting any power.

The unchanged odd frame is

```
A^2-W^3(W+2)(2z+1)^2=1.
```

No sign graph is retained on A. Put a=|A|. Classification at center W+1, with ordinate |W(2z+1)|, gives a=X_l(W+1), with positive l divisible by W. Hence

```
a>=X_W(W+1)>W^W,        |A-1|>4W^3.
```

This argument uses neither a power output's sign nor a binomial interpretation.

4.3 The reused signed anchor

Reuse the derived-label seed pair (U,C), including its original norm and A∗C multiplication gate. No fresh duplicate anchor norm is created. Retain one auxiliary pair (H,J), one comparison pair (S,R), and a comparison center D, with

```
U^2-(A^2-1)C^2=1,
H^2-(A^2-1)J^2=1,
S^2-(D^2-1)R^2=1,
J=C^2(2e+1),
D=1+4J(2d+1),
D=A+Hf,       S=U+Hg,       R=h+4Cq.
```

Every displayed coordinate is an unrestricted integer. The squared cap is

```
C^2=h^2+1+sum_(i not anchor) y_i^2+u^2+v^2+w(w+1).        (1)
```

The ternary form is always nonnegative. Therefore |C|>h, so C is nonzero. The odd auxiliary factor gives J nonzero and C^2 divides J. Consequently |H|>=a and |D|>=3. D is congruent to 1 modulo 4C even when C, J or D is negative.

Classify the anchor and auxiliary pairs at a:

```
|U|=X_i(a), |C|=Y_i(a), |H|=X_l(a), |J|=Y_l(a).
```

Here i,l>0. The inherited squared-divisibility Pell lemma gives i<=l and |C| divides l. Polynomial transfer through D=A modulo H and S=U modulo H gives comparison index j=+/-i modulo 2l. For A<0, the additional factor (-1)^j in X_j(A) is absorbed by the independent abscissa sign. At D=1 modulo 4C, the ordinate recurrence gives j=+/-h modulo 4|C|. Hence i=+/-h modulo 2|C|. Since 1<=i<=|C| and 0<h<|C|, only i=h is possible. Thus

```
|C|=Y_h(a).
```

The proof uses the absolute values of the signed anchor and auxiliary ordinates. In particular neither C>0 nor J>0 is an implicit assumption.

4.4 Pooled signed-index recovery including the anchor

For every other principal pair, (1) gives |y_i|<|C|, hence its classified index satisfies j_i<h<W. For the anchor itself j=h<W. Retain all fifteen terms in the single pooled row, including the anchor term:

```
sum_i (y_i-k_i)^2=(A-1)q0.                                (2)
```

The free quotient q0 has no sign graph. If A>0, reducing the recurrence modulo A-1 yields y_i=epsilon_i j_i. If A<0, a=-A is congruent to -1 modulo A-1 and yields y_i=epsilon_i(-1)^(j_i-1)j_i; at j_i=0 the residue is zero. Thus (A-1) divides the nonnegative integer

```
E=sum_i (r_i-k_i)^2,
```

where r_i is the indicated signed integer index. Since all j_i<W, |k_i|<W and the fifteen pairs are fewer than W,

```
0<=E<4*15*W^2<4W^3<|A-1|.
```

Therefore E=0 and y_i=Y_(k_i)(A) for every pair. Keeping the anchor's pooled term is essential: for an even h and a negative A its correct ordinate is negative. A positive linear cap combined with anchor reuse would not justify this signed-center variant.

4.5 Powers and the acyclic front-sign schedule

For any positive requested index k<W, base 2<=n<W, and signed candidate output |m|<W, put T=2An-n^2-1. Its magnitude exceeds W^W+W. Once y=Y_k(A), the two possible norm abscissas are +/-X_k(A). The correct choice gives m=n^k modulo T. The wrong choice gives m∗n^k=-1 modulo T, by multiplying the two conjugate recurrence expressions; this identity holds for either sign of A. Since |m∗n^k+1|<W^W+1<|T|, the wrong case would imply the impossible equality m∗n^k=-1. The correct congruence similarly gives m=n^k. For base one, the already oriented t pair gives the result directly.

The retained front signs are exactly those of round16 except for A. Initially x>=0, all nine encoded representatives minus one are nonnegative, the factorial remainder and slack are nonnegative, and the exact-binomial output slack is nonnegative. Therefore B=r+s+1>=1, q=B+o+1>=2 and t=16+h0^2>=16. First apply the signed-center power argument to q^t. This orients the t abscissa to X_t(A), not necessarily to a positive number. The same pair gives V0=(F^2+1)^t. The raw ceiling identity FB=V0+s>0 gives F>=1, and then B>=2, q>=3 and y=V0(V0-1)>=F+1. Thus K=2(2+t+x+y+F)>0. All representative indices are positive, and so is the derived-label index. The B^K request gives N, its shared pair gives q=(B H0)^K, and all remaining q or q+1 requests follow. This is the same acyclic order as round16.

Only after the power interpretations are established do the eleven product guards r_i∗s_i>=0 combine with the positive denominator equations r_i+s_i=L_i-1>=0 to give both signs. The factorial fixed point, all range tests and Horn decoding then proceed exactly as in round16. Signed A changes no semantic front value. Private Pell quotient signs need not become natural: if comparison with the older all-natural private representation is wanted, reconstruct its witnesses afresh at a positive center.

4.6 Cofinal semantic-anchor completeness

Begin with a finite refutation and perform the round16 cofinal horizon and beta-CRT construction. It fixes t,F,V0,y,x,B,K,N before the independent representatives. Since t>=16 and F=t!, each d_i=1+F^2/i is odd, so N and N^2 are odd. The representative moduli multiply to N^2.

Choose the ordinary positive representatives in their required residue classes with 0<a_i<=2N^2, prescribing parity by adding N^2 when needed. For the dominant a representative, start with its least nonnegative residue a0 in [0,N^2), and choose a=a0+2N^2 or a0+3N^2 to have parity opposite to b. Then 2N^2<=a<4N^2. The literal monotone derived-label constructor gives h>=2a^2, and its parity is h=1+a+b modulo 2. Thus h is even and exceeds every other requested index, indeed it can be chosen to exceed twice their maximum. The inherited fixed-front bounds are far below N^2. The self-sized radix H0=1+sum representatives is chosen only afterward, so q=(B H0)^K still dominates all required binomial coefficients; no new circular modulus choice is introduced. The factor-of-four increase in the representatives is harmless because the radix uses their actual values rather than an old aggregate ceiling.

Form the actual square frame, choose A=X_W(W+1)>0 and the ordinary odd frame witness, and use canonical principal pairs. Set U=X_h(A), C=Y_h(A). If M=max_(i not anchor) k_i<h, then Y_M(A)<=Y_(h-1)(A). Recurrence gives C>(2A-1)Y_(h-1)(A). As A>W^W, h<W and the number of other pairs is 14,

```
C^2-sum_(i not anchor)Y_(k_i)(A)^2
  >[(2A-1)^2-14]Y_(h-1)(A)^2>h^2+1.
```

Thus the slack in (1) is nonnegative and has a ternary representation. Every term in (2) is divisible by A-1, so its free quotient exists.

To construct the odd auxiliary quotient explicitly, use h even. Then C=Y_h(A) is even and U=X_h(A) is odd. Choose the common index l=hC, and let H=X_l(A), J=Y_l(A). Composition gives J=C∗Y_C(U). The binomial Pell-polynomial expansion gives

`Y_C(U)=C U^(C-1) modulo C^2,`

because U^2-1=(A^2-1)C^2. Thus J/C^2 is an integer congruent to U^(C-1) modulo C, so it is odd. Set e=(J/C^2-1)/2. Also H is odd and the Pell norm gives gcd(H,J)=1, hence gcd(H,8J)= 1. CRT chooses D>=2 with D=A modulo H and D=1+4J modulo 8J. Set d=(D-1-4J)/(8J), S= X_h(D), R=Y_h(D), and fill the three congruence quotients. This supplies every anchor witness without an e sign graph.

The elementary universal representation n=u^2+v^2+w(w+1) for every n>=0 fills all retained guards. The old arithmetic front and every genuine power request are unchanged. Hence each finite refutation extends to a zero of the exact new system.

4.7 Literal compilation and audit boundary

The emitted source records reused anchor coordinates and index, the signed-center flag, squared-anchor flag, odd auxiliary coefficient, every source edit, all frame atoms, every guard and all original power requests. The independent checker does not import the producing compiler. It reconstructs the complete source, checks the exact guard identities and their noncircular sign dependencies, replays Horn factorization and every integral monic restoration, composes integral translations, and independently collects every quartic coefficient. It runs both normally and under Python -O. Deliberate malformed source, frame, guard, factorization, restoration, translation and polynomial controls are rejected.

These finite checks certify the actual finite polynomial and its exact algebraic transformations. They do not replace the unbounded Pell, CRT, factorial, Horn-flow and logical-calibration arguments. Those arguments establish both directions without an assumption of consistency. Negating the solvability statement proves, in ZFC, that this polynomial has no integer zero if and only if Con(ZFC).

5 Recovering nonnegative arithmetic from signed Pell orientation

This proof extends `PROOF_315.md`, whose signed-center frame, squared semantic anchor, odd auxiliary construction, literal compiler and inherited calibration boundary are retained. The changed sign proof below replaces its front-sign schedule. It is not a claim that the old shifted representative coordinates become nonnegative. The proof applies to `orientation_first`, `orientation_separate` and `orientation_groups5`; their centered endpoints are `integer285`, `integer286` and `integer290`.

The exact centered metrics are respectively (variables, rows, monomials, height) = (285,194,6594,128446), (286,195,5770,128446), and (290,199,4949,128446). Each polynomial is a fully materialized degree-four sum of squares over unrestricted integer variables. These are finite achieved constructions, not minimality claims.

5.1 The hypotheses available before any semantic arithmetic

Retain only the factorial remainder and slack scalar graphs r>=0,s>=0, and the exact-binomial output slack scalar graph o>=0. The cofinal horizon remains t=16+h0^2. The literal rows give B=r+s+1>=1 and q=B+o+1>=2. Neither x, the nine shifted representatives, the frame center A, nor any proposed power output is assumed nonnegative.

Let a_i be the actual nine representatives, each equal to its stored coordinate plus one. They are unrestricted at this stage. Let h be the literal derived-label expression, also unrestricted. The square frame still contains every base, exponent and output atom; in particular |h|<W and every requested |k_i|<W. It gives W>=258 and, from the odd frame norm, a=|A|>W^W and |A-1|> 4W^3. These statements use squares of signed atoms, not the postponed signs.

5.2 A signed nominal index in the squared anchor

The exact anchor equations and odd auxiliary multiplier are those in `PROOF_315.md`, but do not presume h>0. The squared cap gives |C|>|h| and C!=0. Classification and squared divisibility give positive indices i,l with |C|=Y_i(a), |J|=Y_l(a), i<=l and |C| dividing l. The two comparison residues imply i=+/-h modulo 2|C|.

If h=0, this is impossible for 1<=i<=|C|. If h!=0, |h|<|C| implies i=|h|. Thus h!=0 and

```
|C|=Y_|h|(a).
```

For every other principal pair the cap gives |y_i|<|C|, hence j_i<|h|<W; the anchor has j=|h|<W. This is the complete, noncircular bound on all classified indices. It does not use any monotonicity property of the signed derived-label expression. In fact |h|=1 is also excluded by |C|>|h|, but that extra fact is not needed until h is oriented.

5.3 Both coordinate signs from bounded errors

Replace ordinate-only pooling by

```
sum_i [(x_i-1)^2+(y_i-k_i)^2]=(A-1)Q,                    (1)
```

including the anchor pair. Q is an unrestricted integer. The separate-pool endpoint instead has one row for all abscissa errors and one for all ordinate errors. The grouped endpoint partitions the fifteen pairs into blocks of five, and uses one abscissa and one ordinate row for each block. The proof below applies to each such row separately.

At A>0, a classified pair x_i=sigma_i X_j(a), y_i=epsilon_i Y_j(a) reduces modulo A-1 to (sigma_i,epsilon_i*j). At A<0, a=-A is congruent to -1 modulo A-1, so it reduces to (sigma_i*(-1)^j,epsilon_i*(-1)^(j-1)*j). The ordinate residue is zero at j=0. Call these small residues s_i and r_i.

Each row therefore makes A-1 divide a nonnegative sum of its errors (s_i-1)^2 and/or (r_i-k_i)^2. Here |s_i-1|<=2, |r_i-k_i|<2W, and the total number of pairs is m=15. The full sum satisfies

```
0<=E<4mW^2+4m<4W^3<|A-1|,
```

where W>=258. Hence every error vanishes. For either sign of A this says exactly

```
x_i=X_(k_i)(A),        y_i=Y_(k_i)(A),
```

with X_(-k)=X_k and Y_(-k)=-Y_k. In particular negative A does not imply that x_i is positive: it is the correct Pell polynomial at the original signed center.

5.4 Negative requested exponents are impossible at a positive base

Let a genuine power request have 2<=n<W, signed requested index |k|<W and signed output |m|<W. Its polynomial modulus is T=2An-n^2-1. Since |A|>W^W,

```
|T|>W^W+W.
```

The canonical signed Pell polynomials satisfy the usual power congruence. If k>=0, it gives m=n^k modulo T. Both quantities have absolute difference below |T|, so m=n^k, including k=0 and m=1.

If k=-j<0, the recurrence at -j is the inverse unit. Multiplication by the positive-index congruence gives

```
m*n^j=1 modulo T.
```

Equivalently gcd(n,T)=1 because T=-1 modulo n, so this inverse is legitimate. Since |m∗n^j-1|<W^W+1<|T|, equality m∗n^j=1 would follow. This is impossible over the integers for n>=2,j>=1. Consequently every such power request itself forces k>=0 and supplies its exact nonnegative power output. No sign of m or of k was used as a premise.

For n=1 and a known nonnegative k, the same canonical congruence directly gives m=1. This handles the possible initial F=0 case without a wrong-abscissa exception, because both coordinates are already oriented by (1).

5.5 The complete acyclic recovery schedule

1. The initial q>=2 and t>=16 interpret q^t. The common t pair is already canonical. Apply its other request with base F^2+1>=1 to obtain V0=(F^2+1)^t>=1.

2. The raw ceiling identity is F∗B=V0+s, with B>=1 and s>=0. Hence F>=1. Also F∗B>=V0>=F^2+1, so B>F and B>=2. Therefore q>=3. The cofinal beta value y=V0(V0-1) satisfies y>F+2; this follows immediately for F=1 from t>=16, and for F>=2 from V0=(F^2+1)^t.

3. The request with base q and exponent x+1 now forces x+1>=0, so x>=-1. The first endpoint row is

   ```
   x=F+1+(y+1)u0.
   ```

If u0<=-1 then x<=F-y<-2, a contradiction. Thus u0>=0 and x>=F+1>=2. The last endpoint row is

```
x=1+[1+(t+1)y]ut.
```

Its positive coefficient and x>=2 force ut>=1, giving x>=2+(t+1)y. These conclusions precede the factorial interpretation.

4. Every actual independent representative a_i is the exponent of a request at base q+1>=4. Each is therefore nonnegative, and each numerator request is exact, even if a_i=0. The same argument gives h>=0 for the anchor request. Since the anchor already excluded h=0, it is positive. The ordinary monotone syntax interpretation of h is now available.

5. K=2(2+t+x+y+F)>0. The requests at B and B∗H0, where H0=1+sum a_i>=1, give N=B^K and q=(B∗H0)^K. Every remaining request is exact by the same canonical power lemma; all eighteen requested powers have been recovered. The derived t+1 pair is still the unchanged one-step Pell addition from the canonical t pair.

6. Each of the eleven delayed remainder pairs satisfies r_i+s_i=L_i-1>=0, because its denominator L_i is now a positive exact q-power. Its retained product graph r_i∗s_i>=0 implies both signs. The corresponding quotient signs follow from their bounded remainders and positive divisors. The output slack o was already retained. These deductions do not justify an earlier power step; they are explicitly later.

7. The ceiling gives V0<=B∗F<B^2. Since K>=2 and H0>=1, q>=B^2>V0. Thus the unchanged factorial digit is exact: B=binom(F^2+t,t). The inherited square-output fixed-point proof for t>=16 now yields F=t! and identifies B as the product of the pairwise coprime d_i=1+F^2/i.

This order uses no sign graph or arithmetic meaning that it is attempting to establish.

5.6 Zero representatives and the precise semantic boundary

The shifted stored coordinate a_i-1 can now equal -1. Therefore it would be false to claim that an arbitrary zero restores every variable of the old natural source pointwise. Instead we have recovered the actual semantic CRT representatives a_i>=0. These are the upper arguments in the range-binomial conditions and the inputs to the arithmetic source. Nonnegative, not strictly positive, upper arguments are sufficient for the inherited range-decoding proof.

For n=0 and k>=1, binom(0,k)=0. The positive-degree coefficients vanish, while the constant coefficient is one. All congruence and prime-power loss arguments remain valid: no step divides by n or assumes n>=k. The range condition still recovers local parameters in the required nonnegative interval. Their constructor and Horn polynomials are then evaluated on nonnegative local values exactly as before. The precision and unit-weight flow argument are unchanged, so a finite refutation is recovered even if some global representative is zero.

The radix bounds need one explicit repair. Let C0 be the encoded index representative, distinct from the private cap ordinate C. After the factorial interpretation, the retained congruence is

```
B divides C0+F^2+1.
```

Because t>=16, the product B=(F^2+1) product_(i=2)^t(1+F^2/i) is strictly greater than F^2+1. Thus C0=0 is impossible and C0>=1. Consequently H0=1+sum a_i>=2.

For every ordinary upper representative a>=0 and every coefficient needed through a positive lower index k, use

```
binom(a,j)<=max(1,a)^k<=H0^k<q,
```

rather than the invalid bound a^k at a=j=0. For the derived label let n=max(a,b,c,d), and n∗ =max(1,n). The inherited positive-input degree ledger gives h<=(n∗+1)^ell, with ell=4194304. Because C0>=1 is a separate representative, H0>=n+2 and hence n∗+1<=H0. Thus h<=H0^ell. The existing t!, K and ell inequalities still give all required derived-label coefficients below H0^K< =q. No extra sign or nonzero premise is hidden in this repair.

The local source is interpreted on the decoded nonnegative values, not on the old shifted global coordinates. The ordinary C222/refutation calibration therefore applies. This establishes the soundness direction without a claim of pointwise restoration to the old natural private or shifted representation.

5.7 Completeness and exact audit

Completeness is unchanged from `PROOF_315.md`: start with a finite refutation, choose all actual CRT representatives positive, choose the dominant a representative with parity opposite b, obtain an even semantic anchor h, and construct the positive canonical frame and all private witnesses. Every abscissa error X_k(A)-1 and ordinate error Y_k(A)-k is divisible by A-1. Thus (1), its separated version, or every grouped row has an integer quotient. Removing ten sign graphs only removes constraints that these canonical witnesses already satisfied. The squared-cap and odd-auxiliary completeness arguments remain fully constructive.

The independent literal checker reconstructs the precise pool partitions, source deletion boundaries, every remaining sign graph, all power atoms, complete factorization, integral monic projection, translations and collected quartic coefficients. It is run normally and under -O. The independent mathematical review separately checks the unbounded signed-anchor and orientation-first arguments. These finite checks do not prove consistency or replace the inherited logical calibration. Together with that calibration, the two directions establish in ZFC that no integer zero exists exactly when Con(ZFC).

6 Aligning the amplification exponent with a strict beta quotient range

This is the final semantic delta to `ORIENTATION_FIRST_PROOF.md` and its fully stated signed-anchor dependency `PROOF_315.md`. The primary released endpoints are `strict_dimension_centered`, `strict_grouped_centered` and `strict_sparse_centered`. Their exact (variables, rows, degree, monomials, height) counts are (282,192,4,5491,128446), (287,197,4,4313,128446), and (309,219,4,3922,128446). The 283-variable separated-pool point is an additional measured diagnostic, not needed to define these three equations.

6.1 Exact changes to the arithmetic interface

Replace K=2(2+t+x+y+F) by K=x. Tighten only the two beta quotient ranges for u and vq: replace their binomial lower index x+1 by x, so the decoded quotients lie in 0<=u,vq<x. The existing q^(x+1) denominator request becomes q^x. All other range lower indices, every numerator, and all other arithmetic equations remain unchanged. The N=B^K and q=(B∗H0)^K requests now use the already existing x-index principal pair. The old independent K pair and its duplicate A∗Y norm wire are aliased to that pair; their duplicate norm and multiplication definition are deleted. There are fourteen, not fifteen, independent principal pairs, while the exact power-request count remains eighteen.

The literal source transformation records each alias and both deleted rows. The independent checker reconstructs these changes from the old and new norm identities, not from a free-form assertion. The raw power, seed, index, amplification, radix and affected range metadata are updated together. The ordinary bound R=2+t+x+y+F remains a mathematical majorant; it is no longer the amplification exponent.

6.2 Noncircular power and endpoint recovery

All initial frame and squared-anchor arguments in the orientation-first proof still apply. The frame uses the actual new exponent x, not an obsolete x+1 or K atom. All classified signed indices are bounded before any arithmetic signs are recovered. Every pooled error proof may use the conservative upper count m<=15, although the actual count is fourteen.

Initially t>=16, B>=1 and q>=2 are supplied exactly as before. Canonical orientation gives V0=(F^2+1)^t. The raw ceiling gives F>=1, B>=2, q>=3, and y=V0(V0-1)>F+2. The denominator request now has base q>=3 and exponent x. The negative-index power lemma therefore forces x>=0 and gives q^x exactly.

From x=F+1+(y+1)u0, a negative u0 would give x<=F-y<0, so u0>=0 and x>=F+1>=2. The last endpoint x=1+[1+(t+1)y]ut then gives ut>=1 and

```
x>=2+(t+1)y.
```

Thus K=x>=2 is known before using the B^K and (B∗H0)^K requests. They give N=B^x and q=(B∗H0)^x. The remaining canonical-power and postponed interval-sign arguments are unchanged. In particular q>=B^2>V0 still follows from the raw ceiling, so the exact factorial digit and fixed point give F=t! without any circular use of the new majorant.

6.3 Amplification, precision and radix bounds

After F=t!, the endpoint bound and y>=F+1 give

```
x-(2+t+y+F)>=t(y-1)-F>0.
```

Hence R=2+t+x+y+F<2x. Also x>=F+1>=16!+1>2^32. Let E=8388608=2^23, the unchanged checked local-polynomial majorant exponent. For every real x>=2^32,

```
E log_2(2x)<x.
```

For an elementary discrete proof, if 2^(n-1)<x<=2^n with n>=33, then E log_2(2x)<=2^23(n+1)<=2^(n-1)<x. The middle inequality holds at n=33 and is preserved when n increases because its right side doubles while n+1 increases by one. At x=2^32 directly E∗33<2^32. Therefore the exact inherited local bound gives

```
L<R^E<(2x)^E<2^x=2^K.
```

This is the same strict amplification inequality required for residue extraction and Horn-flow decoding, now with K=x.

Every range lower index k is at most K. The two changed ones equal x. The others are t, t+1 and y+1, all less than x by x>=2+(t+1)y and y>=1. Consequently the original precision 2K and loss at most k-1<K still suffice. No new divisibility assumption or larger CRT precision is needed.

The self-radix argument for zero representatives is exactly the repaired argument in `ORIENTATION_FIRST_PROOF.md`: the encoded index is positive after the factorial interpretation, H0>=2, and ordinary coefficients are bounded using max(1,a). The derived-label estimate h<=H0^ell also remains valid. The retained factorial inequality ell(t+1)<2F, with ell=4194304, now combines with x>=2+(t+1)y>2F to give ell(t+1)<K. Thus every required derived-label coefficient still lies below H0^K<=q.

6.4 Constructive completeness for the strict ranges

Begin with the same finite refutation, cofinal t, F=t!, beta base y and beta-CRT code x. The first and last endpoint equations supply x>0. For each genuine beta step the canonical quotients have the form

```
u=(x-S)/(1+(i+1)y),
vq=(x-Sp)/(1+(i+2)y),
```

with S,Sp>=0, i>=0 and y>=1. Both denominators are at least two. Thus 0<=u,vq<=x/2<x. This proves that every canonical refutation witness satisfies the newly tightened quotient ranges; it is not merely an upper-bound heuristic.

Choose N=B^x, then choose the CRT representatives at the same precision N^2. Ordinary positive representatives can be chosen with 0<a_i<=2N^2 and prescribed parity. For the dominant a representative, take its least nonnegative residue a0 in [0,N^2) and choose a=a0+2N^2 or a0+3N^2 with parity opposite b. Then 2N^2<=a<4N^2. The literal constructors I(a,b)=2((a+b)^2+a), A(a,b)=2((a+b)^2+3a+4b+3), and the even outer codes give

```
h>=I(a,b)>=2a^2,       h=1+a+b modulo 2.
```

These identities are documented in the copied, hash-bound `constructor_sources/shell_source.py` and `constructor_sources/build_derived_label_range.py`. Thus h is even and dominates every other requested index. All fixed front indices lie below N^2: x,t,y+1 follow from the endpoint bound and N=B^x; F^2+t<N^2 follows from F=t!>=t, B>=F+1 and x>=2: already B^2>F^2+t, and N^2=B^(2x)>B^2. The remaining representatives are below 4N^2, whereas h>=8N^4. The full squared-cap slack, canonical frame and odd-auxiliary CRT construction from `PROOF_315.md` therefore apply verbatim. The complete eighteen-request power interface has fresh valid witnesses at these semantic values.

6.5 Sparse coordinate residues

The sparse endpoint replaces pooled squared-error divisibilities by individual rows

```
x_i-1=(A-1)s_i,       y_i-k_i=(A-1)r_i
```

for all fourteen pairs, including the anchor. The small abscissa residue has absolute value at most two, and the small ordinate-index residue has absolute value below 2W. Both are below |A-1|. Thus they vanish individually and give the same canonical signed-index conclusion. Conversely each canonical pair has the required integer quotients. No scalar sign graph is reintroduced. All other proof stages are unchanged.

6.6 Conclusion and verification boundary

Soundness recovers a semantic front over nonnegative actual representatives and decodes a finite refutation. Completeness builds witnesses from any finite refutation with the strict beta ranges and the chosen even dominant semantic anchor. Composing these directions with the inherited C222/ZFC-refutation calibration proves, in ZFC, that each displayed quartic has no integer zero exactly when Con(ZFC).

The source, guarded source, factorization certificate, integral monic projection, translation and full quartic are all materialized. Normal and -O independent checks verify the complete finite identities and metrics; separate mathematical review checks the ordinary unbounded proof. No global minimum, new formal endpoint, or consistency proof is asserted.

7 Linear cofactor saturation in eleven unrestricted integers

7.1 Result and mathematical boundary

The principal equation below has eleven unrestricted integer variables, exact total degree 328, and 130 live binary arithmetic operations. Its free-scale direct and simple-center companions have degree/cost pairs 352/125 and 340/129. A weak-scale direct equation gives 398/121. These improve the inspected incoming eleven-integer frontier; they are not global minimum claims. The other weak-scale variants, 382/125 and 366/126, are dominated by 352/125 and are retained only as comparisons.

The new arithmetic device places the factors A and Dm inside a linear free cofactor B=r A Dm. A single square-product condition then recovers the square of B, the strong Pell norm, and the auxiliary norm. There is no separate square condition on r or Dm. The previously known main-radicand saturation used a squared free multiplier and an extra factor A; the new construction lowers both the strong norm and the auxiliary coordinate. An exact sharp center removes two further powers of A. The guard uses the round16 modulo-sixteen nonsquare argument, extended here to the free-scale budget.

Every unbounded implication needed for the new transformation is proved below. The signed Pell/rank, intersective-quintic, supported-modulus interpolation and native/compiler/refutation calibration are explicitly retained ordinary mathematical dependencies. Their relevant hypotheses are restored before use. No new proof of consistency, encoded object-ZFC derivation or end-to-end proof-assistant endpoint is asserted.

7.2 1. Complete ordinary polynomials

All eleven independent variables j,C,z,h,r,v,w,T,c,n,s range over Z. The six ports beta,K0,MC,MF,cone_A,cone_B are fixed numerals, with the complete compressed little-endian payloads and finite reconstruction recipe in coefficients/. They are not additional unknowns. In particular beta=2^1064921-1, cone_A=2^1113, cone_B=2^666104, and MF is already shifted by beta.

For the free-scale family define the following polynomial abbreviations:

```
J=6j+1, q=beta J+1, W=12w+4, X=qW, Y=v,
E=XY, a=E+Y, A=a+2, H=4a+3, Delta=a^2+H=A^2-1,
F=(K0+W)C-z(q-1),
```

```
R=[q(q-F)-C](q^2-1)+(MC+q MF)J,
k=R+1+hE, eta=c-Yk,
Derr=(cone_A F-q)^2+(cone_B C-q)^2,
B0=k^2(q^2-Derr)-4q^2 eta^2,
Budget=Y^2 B0-k^2 q^8,
D=Budget-4, ell=XY^2, Dt=1+ell(ell+1)k^2,
Dg=1+D Dt s^2, P=Dt Dg,
Dm=1+Delta c^2,
B=r A Dm, U=r c^2 Dm,
I2=c^4 B=A c^2 U,
Df=1+Delta I2, Q=Delta^2 I2, delta=Q-1,
V=Df(c-R+4A U T)-c, Cy=QV^2-1, Da=delta Cy,
K=X(3c-2X)+2,
M=2q^3, Dc=HM, S=srH M^5,
zc=Dc T+M K+H(v-q^3),
Ac=(zc^2+3Dc^2)(zc^3-19Dc^3).
```

The sharp equation uses lambda=(2Delta+1)U. The simple-center equation uses lambda=2A^ 2U. The direct equation uses lambda=0. In every case put

```
Z=S(n+lambda V)-Ac,
pa=delta Z^2, pb=delta S^2 P, pc=4S^2 B Df Cy,
N=(pa+pb-pc)^2-4pa pb.                                  (1)
```

Equation N=0 is the complete integer polynomial. All symbols are abbreviations, not quantified variables. There are no divisions, radicals or variable exponents in its DAG.

For the weak-scale family use Y=qv, Budget=B0, M=2q^2 and zc=HM T+M K+H(v-q^2). All other formulas remain the same.

The literal implementation shares P=Dt+(Budget-4)(Dt s)^2, L=E(Yk), and

```
Budget=(Yk)^2(q^2-Derr)-q^2[4(Y eta)^2+(kq^3)^2]
```

in the free-scale family. It shares Delta c^2 between Dm and Df. These are identities, not semantic replacements. The three choices of center are integral polynomial shears of n and preserve integer-zero existence with explicit inverse shears.

7.3 2. Unconditional arithmetic and the guard

The authentic affine restrictions give q even, q=2 modulo3, |q|>=beta-1>=16, X=0 modulo8, X= 2 modulo3 and |X|>=4|q|. H is odd and nonzero. If Y=0, a=0, A=2 and Delta=3. Otherwise |a|> =|X|-1>3. Thus A!=0, Delta>1 and Dm>=1 on every integer assignment. Also a=0 modulo3, hence A=2 modulo3 and Delta=0 modulo3.

Dt>=1 since ell(ell+1)>=0. Since ell is a multiple of8, Dt=1 modulo8. Put ucone=cone_A F and vcone=cone_B C. Both are multiples of16. Expansion gives

```
B0+(qk)^2
  =k^2[2q(ucone+vcone)-ucone^2-vcone^2]-4q^2 eta^2
  =0 modulo16.
```

Consequently B0=0 or12 modulo16. In the free family q^8 is divisible by16, so Budget= -(qkY)^2 modulo16, again 0 or12. In the weak family Budget=B0. Hence, for both families,

```
D=8 or12 modulo16.                                      (2)
```

D is nonzero, divisible by4, and never an integer square. Dg=1 modulo4, P=1 modulo4, and neither vanishes. These facts do not assume positivity or Pell solvability.

Q cannot equal1 because it is divisible by Delta^2>1. Df cannot vanish because Df=1 modulo Delta. Thus delta and Cy never vanish. In addition Q is even and delta odd. To see this for all A,c,r, if A is odd then Delta is even; if A is even, Delta is odd, and c^4 A Dm is even. No parity restriction on Y is used.

The rational polynomial pi(t)=(t^2+3)(t^3-19) has no rational zero. Dc!=0, and Ac=Dc^5 pi(zc/Dc), so Ac!=0. At S=0, equation (1) becomes delta^2 Ac^4!=0. Every zero has S!=0 and thus s,r!=0. Consequently B=r A Dm is nonzero.

Da is strictly positive without assuming the sign of B. If c=0 then Q=0 and Da=1. If c!=0 and B<0 then Q<0, so delta and Cy are both negative. If c!=0 and B>0 then Q>1 and Df=1+ Delta c^4 B>|c|+1. Writing V=Df b-c for an integer b, b=0 gives |V|=|c|>=1, and b!=0 gives |V|>=Df-|c|>1. Therefore Cy>0. Finally, B Df is positive whenever c!=0, because B and Df have the same sign. The only possible negative case for Cstar=B Df Da is c=0 with B<0.

7.4 3. Two squareclasses and the linear cofactor

Over a field containing the relevant complex square roots, equation (1) is delta^2 times the four-factor norm with radical terms

```
2S sqrt(B Df Cy/delta),   S sqrt(P).
```

The radicand B Df Cy/delta equals Cstar/delta^2. Thus a zero gives a rational signed sum of 2sqrt(Cstar) and delta sqrt(P), after division by nonzero S and multiplication by delta.

Neither Cstar nor P is zero. If exactly one is negative, its nonzero imaginary term cannot cancel. If both are negative, the preceding section gives c=0, delta=-1 and Cstar=B. Cancellation then requires 4B=P, impossible since P=1 modulo4. Thus Cstar>0 and P>0.

Both are integer squares. Indeed, write each positive integer as a square times a positive squarefree integer. Different nontrivial squarefree radicals are independent over Q. If both terms have the same nontrivial squarefree part, the coefficient of 2sqrt(Cstar) is even, while that of delta sqrt(P) is odd: P is odd and delta is odd. They cannot cancel. If either nontrivial class occurs alone it cannot be rational either. Therefore both squareclasses are trivial.

The three nonzero integer factors B,Df,Da are pairwise coprime. Since Q is a multiple of B and Df=1 modulo B,

```
gcd(B,Df)=1, Da=1 modulo B.
```

Modulo Df one has Q=-Delta, delta=-A^2, V=-c, Cy=-Dm and hence Da=A^2 Dm. Since B=r A Dm, Df=1 modulo A and modulo Dm. Therefore gcd(Df,Da)=1 as well. These statements remain valid with negative B or Df.

Since B Df Da is a positive square and Da>0, the absolute values of B,Df,Da are individually squares and B,Df have the same sign. But Df=1 modulo3, while a negative integer square is 0 or2 modulo3, never1. Thus B and Df are positive. Choose positive integers u,f,e with

```
B=u^2, Df=f^2, Da=e^2.                                    (3)
```

Notice carefully what has and has not been concluded: B=r A Dm is a square, not necessarily r or Dm individually. The proof needs only B.

Since gcd(Dt,Dg)=1 and Dt>=1, positivity and the square condition on P give Dt=tau^2 and Dg=g^2 with positive tau,g. Because s!=0, Budget<=3 would imply Dg=1+D Dt s^2<=0, impossible. Thus Budget>=4, strengthened by (2) to Budget>=12.

For the free family this gives k!=0, Y!=0 and B0>0. As B0<=k^2q^2,

```
Y^2 B0>k^2 q^8 implies |Y|>|q|^3.                         (4)
```

For both families the standard centered disk consequences of B0>0 give Derr<q^2, |c/k-Y|<1/ 2, and the common nonzero sign of q,F,C,q-F-C. In the free family (4) already excludes c=0. The inherited packing bounds are

```
|q|(|q|^2-1)<R<|q|^4-2.                                   (5)
```

7.5 4. Integral auxiliary root and carrier divisibility

Define proof-only integers

```
I=c^2 u, T_old=uT.
```

By (3), f^2-Delta I^2=1, Q=(Delta I)^2, and

```
V=f^2(c-R+4I T_old)-c,                                    (6)
```

because A U=c^2 B=c^2u^2. In particular c^2|I. When c!=0, I!=0 and |I|>=c^2. The rational y=e/delta satisfies

```
QV^2-(Q-1)y^2=1.                                          (7)
```

If c=0, Q=0, e=1, delta=-1 and y=-1 is integral. Otherwise Q>1. The retained adjacent-coefficient denominator descent makes y integral: if its reduced denominator is b then b^2 divides Q-1. The integral unimodular transformation

```
V'=(2Q-1)V-2(Q-1)y,
y'=(2Q-1)y-2QV
```

keeps V' integral, preserves that denominator, and after signs reduces |V| until |V|=1. Indeed V>1 implies V<y<QV/(Q-1), hence -V<V'<V, and the norm excludes V'=0. At |V|=1 one has |y|=1, forcing b=1.

The normalized roots in the four-factor equation are now the integers 2ufy and tau g. Hence Z/S is an integer. The center lambda V is an integer polynomial, so

```
S divides Ac.                                             (8)
```

This conclusion also holds before c!=0 in the weak family, avoiding circularity in its scale restoration.

7.6 5. Scale restoration and exact native source recovery

The free-scale carrier and strengthened budget are inherited from source/FREE_SCALE_PROOF.md, with the fifth-power strengthening retained in source/INCOMING_MAIN_SATURATION_PROOF.md. Here are the complete relevant steps.

Since M^5 divides S, it divides Ac. If p^e exactly divides M and v_p(zc)=b<e, the zc^5 term in

```
Ac=zc^5+3Dc^2 zc^3-19Dc^3 zc^2-57Dc^5
```

has valuation 5b, strictly less than every other term. This contradicts p^(5e)|Ac. Thus M|zc and

```
M | H(v-q^3)                                  (free family).
```

Its 2-primary part, with H odd, gives Y=v even. Also H|Ac and zc=MK modulo H give H|(MK)^5. Put Kq=X(X+2ac)-1. The identity K=HXc-2Kq and odd H yield

```
H | q^15 Kq^5.                                            (9)
```

The omitted main root is recovered before signed rank is invoked. Put Babs=|A|, Cabs=|c|, Iabs=|I|. We have Babs>=3, 0<Cabs<=Iabs, and f^2-(Babs^2-1)Iabs^2=1. Write f=chi_Babs(m), Iabs=psi_Babs(m); then f>2Iabs. Put rho=(Babs^2-1)Iabs. The auxiliary norm classifies rho|V|=chi_rho(nu), |y|=psi_rho(nu); nu is odd because an even-index abscissa is plus or minus1 modulo rho. The odd quotient-polynomial identity and V=-c modulo f give psi_Babs(nu)=plus or minus Cabs modulo f. Fold nu modulo2m and reflect to 0<=p<=m. Since psi_Babs(p)+Cabs<=2Iabs<f, this congruence cannot wrap and forces Cabs=psi_Babs(p), p>=1. This is the absolute folding argument retained in source/signed-quadratic-proof.md Section2 and source/SELECTIVE_FUSION_PARENT.md Section3. It does not require an odd multiplier, a sign of A or c, or a main projection. In particular Dm has now been shown to be an integer square, rather than assumed to be one.

The signed rank theorem now has its complete pre-projection hypotheses: the centered signs and strip, |Y|>|q|^3, even Y, Dt=tau^2, the strong pair (6), c^2|I and I!=0, and V=-c modulo f and V=-R modulo c. The second residue follows immediately from (6). The first-index congruence uses |XY|>3R+1, which follows from (4),(5) and |X|>=4|q|. The theorem does not assume I/c^2 odd and does not require the incoming main radicand Dm to be a separately supplied square. It recovers the main Pell coordinate from the strong and auxiliary norms, as in source/centered11-proof.md Section5.2 and source/INCOMING_LINEAR_MULTIPLIER_PROOF.md Section4.

It yields c=sign(c) psi_A(R), R=2t+1, and

```
|Y|>(|X|-3)^t/2-1, |H|>|X|(|X|-3)^t.
```

The bound (5), with |q|>=16, gives t>=36 (in fact much larger). Write x=|X|, sigma=sign(c), and Pi=(sigma X-2^R)(2^R sigma X+1). The inherited signed Pell projector says 2^R Kq=Pi modulo H. Equation (9) gives H|q^15 Pi^5. Each factor of Pi is less in absolute value than 2^(R+1)x, so

```
|q|^15 |Pi|^5<2^(20t-10)x^25,
|H|>x^(t+1)/2^t.
```

Since the actual x>=2^64 and 64(t-24)-(21t-10)=43t-1526>=22, the divisible integer has absolute value below |H|. Thus Pi=0. Its second factor is odd and nonzero, so sigma X=2^R.

Now q|X makes |q| a power of two. H is odd, so gcd(H,M)=1. The early carrier divisibility finally gives M|v-q^3, restoring

```
Y=(2v_old+1)q^3.
```

The exact projector also gives H|Kq and H|K. Since X=2 modulo3 and R odd, sigma=+1 and X=2^R. The retained first-index sign and centered orientation give k,Y,c>0, and its uniform half-binomial theorem gives 0<eta/k<1/4.

For the weak family, q divides Y from the start, H=3 modulo q and gcd(H,2q^2)=1. The same valuation argument immediately restores v=q^2(2v_old+1), so Y=(2v_old+1)q^3, and also H|K^5. This excludes c=0 from B0>0. The same signed rank theorem applies; the ordinary fifth-power projection bound without the extra q^15 factor suffices (already for t>=24). Its source restoration is identical thereafter.

For precision about the final source handoff, the retained nonodd strong-rank lemma obtains the main Pell index p from |c|=psi_|A|(p), |I|=psi_|A|(m): c^2|I implies p|m, c|m/p, then c|m and m>2p. It uses no odd multiplier hypothesis. After orientation its literal native restoration keeps the outer coordinates and constructs fresh I_star=psi_A(Rc), f_star=chi_A(Rc), rho_star=Delta I_star. Since A is even and R,c odd, I_star and rho_star are odd; composition gives c^2|I_star. Hence r_old=(rho_star/c^2-1)/2 is integral. With t=(R-1)/2, take V_star=(-1)^(t+1)Q_t(rho_star^2), y_star=psi_rho_star(R), using the odd quotient Q_t(z^2)=chi_z(2t+1)/z. The exact residues V_star=-c modulo f_star^2 and V_star=-R modulo c produce the

old private coordinates. Together with the recovered positive strip and the square coordinate 2^((R-1)/2), these give an actual old signed Q41 zero. Only this restored zero is passed to the retained source34/native/compiler/finite-syntactic-refutation calibration.

7.7 6. Converse with one fixed strong pair

Start with a finite syntactic ZFC refutation and the retained canonical outer source witness. In the free family set v to the canonical Y; in the weak family set v=q^2(2v_old+1). Keep all outer values. They give A>0 even, Dm=d^2 with d>0, c=psi_A(R)>0, Dt=tau^2, B0>q^2 k^2/4, and Y>X=2^R>4q^3. Therefore the free Budget exceeds64, and the weak Budget=B0 also exceeds64.

By (2), D=Budget-4 is positive and nonsquare. Dt=tau^2 is a nonzero square, so D Dt is positive nonsquare. Ordinary Pell existence supplies s,g>0 with g^2=1+D Dt s^2. Fix these.

Choose ONE positive strong Pell pair I,f with

```
4s A d c^2 | I, f^2-Delta I^2=1.
```

The retained finite matrix-period construction permits this. Define

```
t_strong=I/(A d c^2), u=I/c^2=A d t_strong,
r=A t_strong^2.
```

All are positive integers. Then r A Dm=u^2, exactly B=u^2, and I2=I^2. Keep this pair I,f fixed for every subsequent step. Every prime dividing u s r divides I: this holds separately for A,d,t_strong and s by the chosen divisibility. Thus the supported-modulus interpolation theorem applies to the modulus u s r, not just to a modulus dividing I.

The exact canonical projector gives H|K. Put kappa=K/H. In either family the carrier has

```
zc=HM(T+kappa+v_old),
Ac/S=H^4 pi(T+kappa+v_old)/(sr).
```

The intersective quintic pi has a root modulo every positive integer. Choose a0 modulo sr with pi(a0+kappa+v_old)=0 modulo sr. Apply supported-modulus interpolation to the SAME fixed strong pair with

```
T_old=u a0 modulo u s r.
```

It supplies an integral auxiliary y. Hence T=T_old/u is an integer and T=a0 modulo sr. Formula (6) is then exactly the polynomial auxiliary expression, and Ac/S is integral. Finally set

```
n=Ac/S+2u f y+tau g-lambda V.
```

Every input is integral, and one factor of the four-factor norm vanishes. This proves completeness. The choice order is outer data, guard s/g, one strong pair I/f, u/r, intersective residue, interpolated T_old/y and T, then n. The strong pair is never reselected after a residue is requested.

Soundness and completeness establish the integer-zero equivalence with the retained calibrated source. Composing its explicitly identified ordinary theorem, ZFC proves that universal nonvanishing of each fixed displayed polynomial over Z^11 is equivalent to syntactic Con(ZFC). This does not prove either side true.

7.8 7. Exact degrees and actual leading coefficients

For free scale the important exact degrees are

```
a3, Delta6, k5, B0_14, Budget18, Dt18, P56,
Dm8, B12, U11, I2_16, Df22, Q28, V37, Cy102,
M3, Dc6, S20, Ac35, lambda17, Z74 (sharp/simple).
```

For weak scale they are

```
a4, Delta8, k5, Budget16, Dt22, P62,
Dm10, B15, U13, I2_19, Df27, Q35, V45, Cy125,
M2, Dc6, S16, Ac35, lambda21, Z82 (sharp/simple).
```

The cofactor relation B Df=B+Delta A^2 U^2 gives the exact center identities

```
lambda^2-4B Df=U^2-4B                    (sharp),
lambda^2-4B Df=4A^2 U^2-4B               (simple).
```

Put W0=Sn-Ac and denote the appropriate right side by diff. The complete cancellation identity is

```
pa+pb-pc = delta W0^2+2delta S lambda V W0+delta S^2P
          +S^2{[delta diff-4B Df]V^2+4B Df}.       (10)
```

The unique highest term in the sharp inner bracket is S^2 Q U^2 V^2, with degree164/free or183/weak. In the simple center it is 4S^2 Q A^2 U^2 V^2, with degree170/free or191/weak. The correction 4pa pb has degree300/free or328/weak, strictly below the respective squared-inner degrees. For the direct form pc uniquely dominates, of degree176/free or199/weak. This proves degrees328,340,352 and366,382,398.

For entire leading forms put q0=6 beta j, X0=12q0w, Y0=v/free or q0v/weak, a0=X0Y0, and e=15/free or10/weak. Then

```
S0=128sr a0 q0^e, B0_top=r a0^3 c^2,
U0=r a0^2 c^4, Df0=r a0^5 c^6,
Q0=r a0^7 c^6, V0=4r^2 a0^8 c^10 T.
```

The entire highest homogeneous form is the following single monomial:

```
sharp:  2^36 s^4 r^18 c^68 T^4 q0^(4e) a0^58,
simple: 2^40 s^4 r^18 c^68 T^4 q0^(4e) a0^62,
direct: 2^40 s^4 r^18 c^68 T^4 q0^(4e) a0^66.
```

In particular the actual degree328 coefficient is

```
[j^118 w^58 v^58 s^4 r^18 c^68 T^4] N
     =2^270 3^176 beta^118 >0.
```

The direct free coefficient is 2^298 3^192 beta^126 at j^126 w^66 v^66 s^4 r^18 c^68 T^4. The weak direct coefficient is 2^344 3^238 beta^172 at j^172 w^66 v^66 s^4 r^18 c^68 T^4. Every exponent sum equals its stated degree. These are exact whole-polynomial proofs at the authentic coefficient tuple, not generic coefficients or numerical line samples.

7.9 8. Portable checks and inherited dependencies

build.py produces the six explicit live arithmetic DAGs. check.py does not import the producer or its circuit module. It independently binds every local polynomial register and live arithmetic gate, expands all four conjugate factors, proves the center and cofactor remainder identities, authenticates all six decompressed coefficient payloads and reconstructs their complete finite recipe, recounts topology/liveness/operations, and propagates a nonzero actual leading coefficient through every node. The only cancellation cut is (10), checked as a free-ring identity and with a unique highest weighted term. The complete per-node attained-leading ledger is emitted alongside the summaries. Coefficient bytes are authenticated, and altered literal, variable, degree and selected formula controls are rejected by explicit runtime checks, including under python -O.

Finite tests are regression controls only. They do not replace the unbounded Pell, interpolation, intersectivity or logical calibration arguments. The full source directory retains those ordinary dependencies, including the nonodd signed-rank argument, the powered saturation interpolation proof, the incoming free-scale proof, and the incoming linear-multiplier proof. SOURCE_PINS.json authenticates their actual bytes. The coefficients, recipe, provenance and license notices are included for portable replay.

8 Height one compilation and its exact witness maps

8.1 The three fixed height one equations

Every row in the following table is an equation over unrestricted integers, with exact total degree four, coefficient height one, individual degree at most four, and constant coefficient one. The actual polynomial is the complete collected coefficient list in the named endpoint directory; the adjacent text file writes every monomial literally.

Endpoint	Variables	Monomials	Source variables	Hubs
compact793	793	51,982	287	16
balanced806	806	23,719	287	48
sparse994	994	11,744	309	192

The first two use the strict grouped 287-variable source, with 197 source rows. The fixed affine refactorization adds 39 coordinates and 39 monic rows, replacing 40 affine-square occurrences and sharing one definition. Coefficient normalization then gives 388 logical coordinates and 298 rows, with 1,284 signed term occurrences. The 793-variable endpoint has 712 factor-alias coordinates, 61 equality dummies and 20 free flows; it uses 36 quartic pair penalties. The 806-variable endpoint has 711 factor-alias coordinates, 48 equality dummies and 47 free flows; it uses 31 quartic pair penalties. Unit aliases are already included in the factor-alias counts, not additional coordinates.

The third endpoint uses the strict sparse 309-variable source with 219 rows. The affine refactorization again adds 39 coordinates and 39 rows. Normalization gives 403 logical coordinates and 313 rows, with 1,264 signed occurrences. The final polynomial has 745 factor aliases, 58 equality dummies and 191 free flows, and uses 29 quartic pair penalties.

Against the inspected incoming height-one equation with 2,215 variables and 30,468 monomials, the balanced and sparse endpoints improve both counts. The 793-variable point trades a larger support for fewer variables. On the identical incoming 442-variable source with 128 hubs, the incidence-local compiler and quartic pairs give 1,254 variables and 27,714 monomials, compared with that incoming 2,215/30,468 result. This matched comparison uses no affine refactorization and therefore isolates the graph and alias changes from improvements to the calibrated source.

The source JSON, transformation JSON and endpoint hashes are listed in `ENDPOINTS.json` and `SOURCE_BINDINGS.json`. There are no unspecified coefficients, source-dependent parameters, or hidden extra unknowns in any final polynomial. The full mathematical proof follows; its source consistency calibration is the enclosing direct-source proof.

8.2 The result and its scope

This construction compiles an explicitly given nonempty quadratic system over $\mathbb{Z}$ into a nonnegative polynomial of degree at most four and coefficient height one. Source rows are collected and nonzero, and every source coordinate occurs; the supplied systems satisfy these conditions. Its new step is to contract aliases using the *actual incidence energy*, rather than protecting every possible pair of factor monomials. Two further improvements factor affine squares before coefficient normalization and replace certain three-coordinate equality cycles by a two-coordinate quartic. The supplied coefficient files, not an unexpanded circuit alone, define the final polynomials.

The correspondence proved below is a bijection between integer zero sets. It is unconditional finite algebra. Applying it to the supplied calibrated source establishes the same ZFC-provable

equivalence between unsolvability and Con(ZFC) as that source. Neither the construction nor its finite checks prove either consistency or universal nonvanishing. No global optimum, literature-wide priority, or new end-to-end formal-kernel consistency calibration is claimed.

8.3 Exact affine refactorization

Let the original system be $R_r(x) = 0$, where each R_r is integral of total degree at most two. Suppose a homogeneous quadratic part is $\epsilon L(x)^2$, where $\epsilon \in \{-1, 1\}$ and L is an integral linear form. Choose an integer k, introduce z, and replace that row by

$$z - L(x) - k = 0, \qquad \epsilon z^2 + R_r(x) - \epsilon(L(x) + k)^2 = 0.$$

The second displayed expression is quadratic because the homogeneous quadratic parts cancel. A previously introduced coordinate for the same affine form may be reused. Every added definition is monic in a fresh coordinate and depends only on original coordinates. Consequently, for every original tuple there is exactly one extension, namely $z = L(x) + k$. Substitution proves the exact pullback identity for every replaced row. Reading the original coordinates is its inverse on zeros. No division, sign assumption, inequality, or domain change is used.

The producer chooses candidate forms by exact integer square tests and coefficient comparison, never by numerical factorization. The independent checker instead reconstructs every affine definition and proves every original-row pullback coefficient by coefficient. Correctness therefore does not depend on the heuristic selecting the sparsest shift.

8.4 Unit coefficient normalization

Write the refactored source coordinates as X. For every needed positive coefficient $c > 1$, and recursively for $\lfloor c/2 \rfloor > 1$, introduce a coordinate t_c and the monic equation

$$t_c - t_{\lfloor c/2 \rfloor} - t_{\lfloor c/2 \rfloor} - (c \bmod 2) = 0, \qquad t_1 := 1.$$

The two equal summands are retained as separate signed occurrences. In increasing order of c, these equations force $t_c = c$. For each needed product cX_iX_j, introduce a shared scale coordinate $s_{c,i}$ with $s_{c,i} - t_cX_i = 0$. Replace a signed source term of absolute coefficient c by, respectively,

$$t_c, \qquad t_cX_i, \qquad s_{c,i}X_j$$

for constant, linear, and quadratic terms. Terms of absolute coefficient one need no change. All resulting rows have the form

$$S_r(y) = \sum_{j=0}^{s_r-1} \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

The new coordinates are uniquely determined by the monic constructors. The normalization checker verifies the topological pivots, their complete coverage, and the exact polynomial pullback of every source row. Thus this step is also a bijection of integer zero sets.

8.5 A shared incidence graph

Fix a positive integer H. Make one hub block of H vertices. For row r, make a row block of $b_r = \lceil s_r/H \rceil$ vertices. Connect the vertices of each block by a tree with a different unrestricted integer flow coordinate on each internal edge. The number of free flow coordinates is exactly

$$F(H) = H - 1 + \sum_r (b_r - 1).$$

For each signed occurrence, add an edge directed from its row block to the hub block, with prescribed value $\epsilon_{rj} M_{rj}$. The explicit placement in the certificate makes these edges distinct; there are no loops or parallel edges. Each M_{rj} is a homogeneous quadratic monomial in factor aliases. A quadratic source term uses aliases of its two logical factors. A linear term uses an alias of its logical factor times a unit alias. A constant term uses two unit aliases. The factors of a square may be the same coordinate.

For each vertex v, let d_v be the outgoing minus incoming edge sum. Define

$$E = \frac{1}{2} \sum_v d_v^2.$$

Before alias contraction, distinct formal edge values have diagonal coefficient one, since every edge has two endpoints. Distinct edges meet in at most one vertex, so their cross coefficient is zero or ± 1. After monomial substitution, coefficients may collide. This is a *coefficient* issue, not a zero-set issue: the exact sum-of-squares identity remains valid under every substitution.

8.6 Incidence local contraction

Every factor coordinate has a logical value: one normalized coordinate y_i, or the distinguished unit value. Contract only coordinates with the same logical value. The producer tries a substitution $b \mapsto a$ and collects *all* affected terms of E. It accepts the substitution only when each resulting coefficient belongs to $\{-1, 0, 1\}$. Opposite signs may cancel; zero coefficients are deleted. No C4-free condition is imposed on the factor graph, and square aliases may merge with other aliases when the exact coefficient test permits it.

This is strictly less restrictive than requiring uniqueness of the product of *every* pair of source monomials. Only a source-edge square or a pair of incident edges contributes to E. A collision between two products not simultaneously present in that support is irrelevant. Even a collision in the support can be harmless if its signed coefficients cancel. The independent checker does not trust the contraction history: it reconstructs the final vertex divergences, expands $\sum_v d_v^2$, and verifies the entire collected polynomial.

Contraction does not change either direction of the intended zero-set map. At every intended zero, aliases with the same logical value were already equal. Conversely, the following equality penalties force all remaining aliases in each logical class to agree. Every factor occurrence is independently checked against its normalized source monomial.

8.7 Equality and unit penalties

For a logical class of $k \geq 3$ distinct aliases, use the cycle energy

$$C(a_1, \ldots, a_k) = \sum_{i=1}^{k} a_i^2 - \sum_{i=1}^{k} a_i a_{i+1}, \qquad a_{k+1} = a_1.$$

Twice this energy is $\sum_i (a_i - a_{i+1})^2$. Its coefficients have height one, and it vanishes precisely when all aliases agree. A singleton requires no penalty. A two-member class can be completed to a three-cycle with one new dummy alias.

There is a better option for a suitable two-member class:

$$\begin{aligned} Q(a, b) &= a^4 + b^4 - a^3 b - a b^3 \\ &= (a - b)^2 (a^2 + ab + b^2). \end{aligned}$$

Since $a^2 + ab + b^2$ is positive except at $(0, 0)$, $Q \geq 0$ and $Q = 0$ if and only if $a = b$. An exact sum-of-squares identity, also used by the checker, is

$$2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2.$$

If neither a^2 nor b^2 occurs as a prescribed edge monomial, none of the four monomials of Q can occur in E: a product producing a^4 needs an a^2 factor, and one producing a^3b needs an a^2 factor; the other cases are symmetric. The lower-degree flow terms cannot collide either. Distinct logical classes are disjoint. Hence this criterion permits replacing the dummy and its cycle by Q, saving one variable and two monomials while preserving height one. The emitted polynomial is checked in full even when this structural sufficient condition is available.

Let $u_1, \dots, u_\ell$, with $\ell \geq 2$, be all unit aliases, adding dummies if needed. Use

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U equals $(u_1 - 1)^2 + \sum_{i<\ell}(u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus $U \geq 0$, with equality precisely when every unit alias is one. Unit, ordinary-alias, and free-flow namespaces are disjoint.

The final polynomial is

$$\boxed{P = E + \sum_{\text{cycle classes}} C + \sum_{\text{pair classes}} Q + U.}$$

It is nonnegative over $\mathbb{R}$. The source-edge products have degree two, so $\deg P \leq 4$. The only constant term is the one in U. Full collected expansion proves that every nonzero coefficient is ± 1, that the exact degree is four for each delivered endpoint, and that every counted variable actually occurs.

8.8 Explicit forward witness map

Given original integer coordinates x, first evaluate all affine refactor definitions, then all constant and scale definitions. Assign every ordinary alias and its dummy the corresponding normalized value, and every unit alias or dummy the value one. All equality and unit penalties then vanish identically.

Root every internal block tree at its first listed vertex. Let b_v be the divergence contributed at v by prescribed edges alone. For an internal tree edge joining a child v to its parent, let $\sigma_v \in \{-1, 1\}$ be that edge's incidence sign at the child. Set its flow to

$$w_v = -\sigma_v \sum_{t \text{ in the subtree rooted at } v} b_t.$$

This is an explicit integer polynomial in the source coordinates. Summing subtree divergences shows that every nonroot divergence is zero. The root divergence is the sum of prescribed divergences over its block. For row block r it is S_r; for the hub block it is $-\sum_r S_r$. Constructor rows pull back to zero, while all other rows pull back to the original residuals. Therefore the full lift L satisfies the universal identity

$$P(L(x)) = \frac{1}{2}\left(\sum_r R_r(x)^2 + \left(\sum_r R_r(x)\right)^2\right).$$

In particular an original zero gives a final integer zero. The displayed identity is *not* generally $P(L(x)) = \sum_r R_r(x)^2$. Its integrality follows either from the integer polynomial construction or from $n^2 \equiv n \pmod 2$.

8.9 Explicit inverse and uniqueness

Suppose $P(z) = 0$ for an integer tuple z. Every summand in the boxed definition is nonnegative, so each is zero. The equality penalties identify all aliases in each logical class, and the unit path forces all unit values to one. Read one listed alias representative from each logical class to obtain y.

Every vertex divergence vanishes because $E = 0$. Summing divergences over row block r cancels its internal edges and yields exactly $S_r(y) = 0$. Topological normalization forces each constant and scale coordinate to its prescribed value; affine refactor definitions force each added affine coordinate. Reading the original coordinates therefore gives $R_r(x) = 0$ for every r.

All remaining coordinates are uniquely determined by this x. The constructor coordinates, aliases, units, and dummies are unique by their definitions. On each internal tree, eliminating a leaf uniquely determines its incident free flow, since its coefficient is ± 1. Continuing to the root gives precisely the subtree formula above. Thus reading the original coordinates is inverse to L on zero sets, proving a bijection, not just equisolvability.

8.10 Reproducibility and evidence boundaries

The literal source equations, affine certificates, graph certificates, and fully collected polynomial files determine every mathematical object. The producer uses exact standard-library integer arithmetic. The independent checkers import no producer, reconstruct all affine source pullbacks, verify every alias occurrence, check the simple incidence graph and every internal tree, and expand twice the final polynomial as an actual sum of squares. Both ordinary Python and `python -O` runs use explicit exceptions rather than removable assertions. Deliberate corruption tests exercise coefficient signs, source hashes, source-row coverage, logical factor bindings, missing edges, alias coverage, unit endpoints, and metrics. Finite off-zero evaluations are supplementary implementation tests only; the preceding algebra establishes the unbounded witness maps.

The incoming shared-incidence construction, coefficient normalization, cycle and unit energies, and their supplied source calibration retain their original attribution. The contribution here is incidence-local coefficient-aware contraction, the collision-safe quartic pair penalty, and their composition with exact affine refactorization and the new source. Comparisons are tied to the inspected archives and exact source hashes.

9 Comparison with the pinned public collection

The source snapshot is ProveIt commit

`1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`.

Its commit time is 04:20:03 UTC on 25 October 2026. It is 50 reachable commits after the preceding pin `f4d8db99e060339f3306a04d507c35bc1d0d68c5`. The placement commit lists the seven wave-14 archives as sources127–133, including the preceding Round16 contribution. Placement in incoming sources is not acceptance of every mathematical claim or integration into active code.

The six other archives were obtained from the exact parent

`fd3e384e92ed4813a5536f877c5cd835ca4c75d2`.

Their full byte streams match the pinned Git blob SHA-1 values and separately recorded SHA-256 values. Internal manifests cover 2842 authenticated entries. The prior Round16 artifact was independently recovered and authenticated. Duplicate names, unsafe paths and CRC failures were checked before using the files. The package keeps the provenance receipts, not copies of every ancestor ZIP or PDF.

9.1 Natural quartic comparison

Every row in the following table is degree four over independent nonnegative integer coordinates. Rows count quadratic residuals; terms and height are measured after collecting the sum of their squares.

Inspected endpoint	Variables	Rows	Terms	Height
Semantic anchor	234	183	9019	87741134
Semantic anchor dense	233	182	50762	88674690
Staged principal	239	187	20015	84062635
Staged sparse	240	188	12836	84062620
Staged minimum dimension	237	185	69710	133499131
Staged small support	255	203	5743	64214623
One-row Pell	242	190	9308	18049118
Shared index	241	189	30058	85187728
Shared index dense	239	187	387494	269805528
Graph certificates	240	188	25987	90589484
Graph certificates dense	239	187	360097	491869488

The new 230-variable point has fewer variables, rows and terms, and smaller height than the inspected 234-variable semantic anchor. It also improves all four listed measures relative to the staged principal and sparse points. The 229-variable point has slightly more terms than the 234-variable semantic anchor and is therefore not said to dominate it. The small-support 255 point retains its lower support and height, while the one-row 242 point retains its lower coefficient height. These genuine tradeoffs must not be erased by ranking only variable count.

The shared-index and graph natural points listed here are dominated in the measured quantities by the staged 239 point. Their mathematical ideas can nevertheless remain useful. In particular the semantic-anchor cofinal choice and the common-root congruence are retained ingredients, not renamed as new inventions.

9.2 Signed direct quartic comparison

All coordinates in this table are unrestricted integers. These equations are not obtained by silently dropping the domain restrictions of a natural system.

Inspected endpoint	Variables	Rows	Terms	Height
Round16 principal	324	208	5808	128446
Round16 sparse	338	222	4493	128446
Public official principal	329	216	5728	128446
Public grouped	364	227	4574	128446
Signed caps free scales	352	219	9355	15020718
Signed caps guarded	355	220	9370	15020718
One-row signed	378	230	9421	3054220
Graph signed	391	235	10764	80496640
Graph signed contracted	381	225	13179	2445790974704

Even the retained Round16 principal dominates the newly inspected incoming signed cap and graph points. The public 329 point is not dominated by Round16 principal 324, because it saves 80 terms. The sparse 338 point dominates public grouped 364 but retains a real support advantage over several lower-dimension alternatives. The new 282 point improves variables, rows and support over both Round16 principal 324 and public 329, with equal height. The new 287 point improves all three counts over sparse 338, again at equal height; new 309 lowers support still further to 3922 terms. The three new points retain distinct dimension/support tradeoffs. Only comparisons respecting all listed metrics are called domination.

9.3 Height one and small-variable objectives

The incoming shared-incidence height-one example uses 2215 unrestricted integer variables and 30468 collected monomials. The new incidence-local constructions preserve height one and exact quartic degree while offering smaller dimension, with support tradeoffs disclosed in the results table. This is a coefficient-height result, not a claim to have fewer variables than the direct signed

quartics. A coefficient-height-one equation with hundreds of additional variables is not universally simpler. A matched-source control isolates the compiler improvement: the same 442-variable signed source that underlies the incoming 2215/30468 example gives 1254 variables and 27714 terms under the new compiler profile. The further gains at 793/51982,806/23719 and 994/11744 also use the improved 287- and 309-variable sources and selected exact affine refactorizations. The archive retains the matched source and reproduces its comparison polynomial and certificate by hash.

The inspected incoming eleven-integer linear-multiplier points include degree/cost 440/126 and 412/127; the free-scale point is 352/132. The new 352/125 point has no larger degree and fewer operations than each of these, under the reported live-binary-gate convention. The eleven-variable cofactor family is a separate high-degree objective. Its degree/cost pairs are 328/130, 340/129, 352/125 and 398/121. Smaller degree is not automatically smaller arithmetic cost. The preceding Round16 product and two-root points include 966/109, 882/110, 810/112 and 654/117, so these remain relevant low-cost comparisons. Positive-domain ten-variable equations are not mixed into the unrestricted-integer frontier. Expanded scalar support and total coefficient height were not computed and are not assigned artificial zero values in a comparison.

9.4 Attribution and limits

The inherited ingredients include the numerical C222 calibration, ordinary Pell and indexed-Pell constructions, square-radius factorial fixed point, factorial/CRT fusion, self-sized radix, cofinal CRT representatives, semantic-anchor reuse, signed rank and interpolation, source34 numerical tuple, unit-coefficient normalization, shared-incidence energy, cycle equality and unit-path energies. The new proof texts identify which exact transformations are added and which earlier ordinary theorems are used.

The full source collection is at the pinned ProveIt repository snapshot. Original coefficient and calibration pins remain separately recorded. No source-publication or priority claim extends beyond the inspected collection; no table is a literature-wide optimum theorem.

10 Reproduction and evidence boundaries

Extract the archive to a fresh directory and run `python3 verify.py`. Mathematical verification uses the Python standard library only. The driver validates every enumerated byte digest, makes a disposable copy, rebuilds the selected artifacts deterministically, compares the rebuilt mathematical objects to the reference bytes, and runs independent checks in normal and optimized Python modes. Every mathematical condition is enforced by an explicit exception rather than a removable `assert`. The checks require no repository checkout, network access, parent workspace or private helper. A separate report build requires a normal TeX installation and is invoked with `sh build_report.sh`.

The disposable-copy rule is substantive: an inherited natural projection checker writes its own verification receipt. Its output must never modify the immutable release reference between normal and optimized passes. All such calls are directed to fresh copies. The manifest excludes only its own bytes and explicitly named replay receipts; the structure checker rejects undeclared files and unsafe paths. Ancestor PDFs, nested ZIP files, exploratory outputs, private notes and acquisition helpers are not part of this release.

The release supplies four different kinds of evidence, with different purposes:

1. *Authentication* binds incoming archives and source files to exact public Git objects and full byte hashes. It says nothing by itself about their mathematics.

2. *Finite exact algebra* verifies every coefficient identity, literal emitted row, monic restoration, translation, graph incidence, numerical port, degree certificate and corruption-control rejection that the checker claims.

3. *Unbounded ordinary proofs* establish the Pell classifications, cofinal choices, inequalities, interpolation and both witness directions. A bounded experiment is not used in place of one of these arguments.

4. *Logical calibration* is the identified inherited ordinary numerical-calculus or native-compiler correspondence with finite syntactic refutations. Retained Lean interfaces identify the boundary; merely including them does not certify the new arithmetic in a proof assistant.

The full proof and exact artifact closure are necessary together. A successful replay is not a proof that ZFC is consistent, and does not decide whether the displayed polynomials have a zero.

10.1 Portable proof closure

The direct arithmetic closure includes `calculus.tex`, `primitives.tex`, `crt.tex`, `direct_arithmetic.tex` and `calibration/inherited_onestack_section.tex`, together with their literal numerical source and calibration interfaces. The complete constructor, derived-label, rank, signed-source and new composition proofs are retained under the component source and proof directories. Scalar source34 coefficient construction, signed Pell folding, nonodd rank restoration, fixed-pair supported-modulus interpolation and compiler/refutation sources are retained separately. The height-one theorem is unconditional as a transformation of an explicit integral quadratic system; its consistency interpretation comes only after composing the calibrated source supplied with it.

All local source names and their hashes are collected in the manifests and component closure maps. Historical copies may refer to their original archive layout; the active release maps identify where their required material is now stored. No external helper URL is a runtime dependency.

10.2 Report fidelity

The PDF is built in three TeX passes, rendered page by page, and checked against a fresh three-pass build for raster identity. The final approved PDF is copied byte for byte into the ZIP. Its editable TeX sources, the proof-rendering script and the independent character-fidelity checker accompany it. Rendering receipts describe presentation checks only; they are not mathematical proof objects.

11 Unfinished directions

These are research questions, not results of this release.

1. Can a smaller signed anchor preserve the acyclic orientation proof? Any deletion needs a proof for the actual emitted rows, not only canonical witnesses.

2. Can cofinal choices lower natural dimension without the large height cost of aggressive monic substitutions? Domain-specific objectives must remain separate.

3. Can incidence-local contraction, affine factorization and hub partition be optimized jointly? The finite search proves no global optimum.

4. Are there better collision-safe equality polynomials that preserve a complete inverse witness map while improving the height-one dimension/support frontier?

5. Can eleven-integer cofactor degree or cost be reduced without changing the authentic numerical tuple or the required interpolation modulus?

6. Can the ordinary arithmetic and inherited logical calibration be joined into a new end-to-end kernel-checked endpoint? This release does not complete that formalization.